

MTU_ValueService **Technical Documentation**

**Technical Specification
for Serial Data Communication
between MTU's
Monitoring and Control System
MCS-5 and External Systems**

Documentation Part 1

Structure and function

E531652/03E



MTU assuring you:

- **ISO 9001 certification**

Quality assurance in design/development, production, installation and service

Printed in Germany

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Il manuale va consultato per evitare anomalie o guasti durante il servizio, per cui va messo a disposizione dall'utente al personale addetto alla manutenzione e alla condotta.

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**Technical Specification
for Serial Data Communication
between
MTU's MCS-5 Monitoring and Control System
and
External Systems**

1 General

The MCS-5 interface protocol is used for transmitting data between the MTU MCS-5 monitoring and control system and external systems.

The external system is connected to the MCS-5 by means of interface couplers. The following interface couplers can be used:

- A communication process station (CPS)
- A serial interface (PIM 1)

The purpose of the interface couplers is to enable data exchange between the serial interface of the external system (RS422 (standard) or RS232 (optional)) and the CAN bus of the MTU MCS-5 monitoring and control system.

The interface couplers are compatible with full duplex operation and provide for conductive decoupling.

2 Data Communication

No initialization is necessary before serial communication is started. No handshaking procedures are used. Serial data communication takes place in full duplex mode. The rules described in this document apply to the transmitting as well as receiving of data.

Serial communication (transmitting and receiving) occurs cyclically in a continuous loop. The cycle time for transmission of all data blocks, as well as the minimum pause intervals which must be observed, are specified by MTU separately for each equipment order.

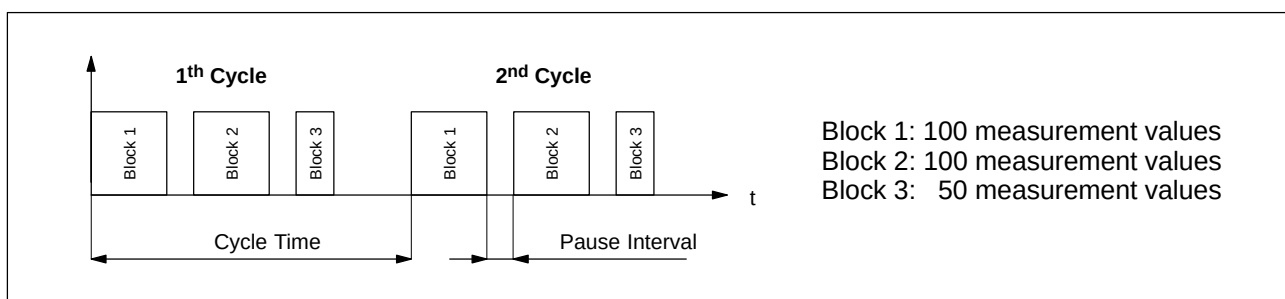


Fig. 1 : Example of one data cycle for transmission of three measurement-value blocks

The sequence of the blocks in the cycle is determined by the specified number of the block. If an alarm occurs, the sequence of the data blocks is changed (see Item 5).

The data stream is monitored by the respective receiving system. If data transmission is interrupted, the receiving system must generate an error message.

The measures to be taken in case of transmission errors are described in Item 6.

Data transmission takes place in blocks; there are three different block types (xxh - number in hexadecimal format):

Block type	Block type identification
Analog measurement value block	03h
Binary measurement value block	02h
Request block	01h

Two Bytes (unsigned, Motorola storage format) are used for identification of the block type.

The data blocks are embedded in a data frame which is constructed identically for each block type.

Start_1	Start_2	Block-No.	Block Type	Frame Size	Measurement value	...	Measurement value	Check-sum	End_1	End_2
1 Byte	1 Byte	2 Byte	2 Byte	4 Byte	X Byte		X Byte	4 Byte	1 Byte	1 Byte

Data block

Tab. 1: Data frame

The start and end of a data frame are identified by the following values:

Start_1 ⇒ FAh
 Start_2 ⇒ F5h
 End_1 ⇒ AFh
 End_2 ⇒ 5Fh

The block number (unsigned, 2-Byte integer value, Motorola storage format) follows the start identifier; block numbers are assigned by MTU specifically for each equipment order. The block number "0" is reserved for the request (see Item 7) and is used by the transmitter as well as by the receiving system.

The entry "Frame Size" (unsigned, 4-Byte integer value, Motorola storage format) contains the number of all Bytes transmitted in the data frame, i.e. the number of Bytes from Start_1 to End_2.

When the MCS-5 receives a data block, the data block contains a maximum of 100 measurement values. If the CPS interface coupler is installed, a maximum of 25 measurement values can be sent. If the PIM interface coupler is installed, a maximum of 100 measurement values can be sent.

Consequently, the external system can send a maximum of 100 measurement values in a data block. If the CPS interface coupler is installed, a maximum of 25 measurement values can be received. If the PIM interface coupler is installed, a maximum of 100 measurement values can be received.

The external system interface should be configured to receive and process 100 measurement values.

The minimum number of measurement values contained in a data block is 1.

Allocation of the transmitted data to measurement points takes place in accordance with an order-related measurement-point list. To ensure trouble-free data communication, it is absolutely necessary that the specified sequence be strictly observed.

A checksum (unsigned, 4-Byte integer value, Motorola storage format) is transmitted following the data block; the checksum is formed by adding all Bytes starting with "Start_1" and ending with the last data Byte before the checksum.

The data frame is concluded by an end identifier.

The end identifier may only be checked after the number of Bytes specified by Frame Size has been received because it is possible that data equivalent to the end identifier may be transmitted as part of analog measurement value blocks within the data block.

3 Analog Measurement Value Block

Start_1	Start_2	Block-No.	Block-Type	Frame Size	Measurement value 1	...	Measurement value X	Check-sum	End_1	End_2
1 Byte	1 Byte	2 Byte	2 Byte	4 Byte	4 Byte		4 Byte	4 Byte	1 Byte	1 Byte

Data block

Tab. 2 : Structure of an analog measurement value block (X ... max. number of measurement values (see Item 2))

Analog measurement data are transmitted as 4-Byte integer values with signed integer values in Motorola storage format (see Tab. 3). The integer value which has the highest-value Byte is the first one transmitted in this storage format.

Byte 4 Highest-value Byte	Byte 3	Byte 2	Byte 1 Lowest-value Byte
00h	00h	3Ah	B8h

Tab. 3 : The analog measurement value "15032" = "3AB8h" in Motorola's format.

Measurement values are to be standardized as stated in the measurement-point list provided by MTU.

"80 00 00 00h" is transmitted in case of a defective sensor.

Missing data are identified by the value 7FFF FFFFh.

Note: "Missing Data" does not represent a critical value. Simply, there is no information about the contents of the measurement point in this case.
Connected external equipment must assume a safe state on receiving a measurement point assigned "Missing Data".

4 Binary Measurement Value Block

Start_1	Start_2	Block-No.	Block-Type	Frame Size	Measurement value 1	...	Measurement value X	Check-sum	End_1	End_2
1 Byte	1 Byte	2 Byte	2 Byte	4 Byte	1 Byte		1 Byte	4 Byte	1 Byte	1 Byte

Data block

Tab. 4 : Structure of a binary measurement value block (X ... max. number of measurement values (see Item 2))

Binary measurement values are transmitted as signed integer values. The binary measurement value can have the following values:

01h	⇒	True
00h	⇒	False
80h	⇒	Defective sensor
7Fh	⇒	Missing data

5 Alarm Messages

Alarm block transmission within a data cycle depends on the interface coupler installed.

As a rule, the data cycle always begins at the same time the alarm blocks are transmitted (the number of alarm blocks depends on the order). Once the alarm blocks have been transmitted, the remaining data blocks are transmitted in the agreed block sequence.

- **Serial interface (PIM 1)**
If an alarm occurs while a data block is being transmitted, the transmission is ended in the correct manner. The data cycle is restarted once the pause time has expired.
- **CPS communication station**
If an alarm occurs while a data block is being transmitted, the transmission is ended in the correct manner. The alarm block which contains the new alarm message is added to the data cycle once the pause time has expired. Once the alarm block has been sent, and provided no further alarm blocks need to be sent, the next data block must be transmitted in the agreed block sequence. The adding of alarm blocks may cause the specified cycle time to overrun. In this instance, the next cycle is started once the pause time has expired. If the data transmission (data blocks with additionally added alarm block) ends before the cycle time has expired, the next cycle will begin either after the remaining cycle time has expired or after the pause time has expired, whichever is longer.

If an alarm becomes active in the time between the end of a data cycle and the start of the next data cycle, the interface couplers react either immediately or after the pause time has expired, as described.